Points: 6.75/17
(39.71%)

Aufgabe 1.

Let R be a dedekind ring, I, J ideals of R.

We define $kgV(I,J) = I \cap J$.

i)

t.s. if $I \neq 0$, $J \neq 0$ then $kgV(I,J) \neq 0$.

proof. By ex.S 3., it holds that $IJ \subseteq I \cap J$, ok
and by assumption there exist $0 \neq a \in I$, $0 \neq b \in J$
thus it follows that $a \cdot b \in I \cap J = kgV(I,J)$ and we can
conclude. ☐

Why is it $\neq 0$?

0.75

ii)

t.s. $I \mid kgV(I,J)$, $J \mid kgV(I,J)$

proof. This follows from Corollary 2.20. and
the fact that $kgV(I,J) = I \cap J \subseteq I, J$. ☐

2

iii) Let $K \subseteq R$ be an ideal with $I \mid K$ and $J \mid K$.

t.s. $kgV(I,J) \mid K$.

proof By corollary 2.20 we get ✓
$I \mid K \;\&\; J \mid K \Leftrightarrow K \subseteq I, J \Leftrightarrow$ ✓ $K \subseteq I \cap J = kgV(I,J)$
$\Leftrightarrow kgV(I,J) \mid K$.
✓
☐

2

Aufgabe 2

Let $I = (2, 1+\sqrt{-3}) \subseteq \mathbb{Z}[\sqrt{-3}]$.

i)

t.s. $I \neq (2)$.

proof It holds that $1+\sqrt{-3} \in I$ ✓ and furthermore it holds that $2 \nmid 1+\sqrt{-3}$ in $\mathbb{Z}[\sqrt{-3}]$. ✓

since $\frac{1+\sqrt{-3}}{2} = \frac{1}{2} + \frac{1}{2}\sqrt{-3} \notin \mathbb{Z}[\sqrt{-3}]$. ✓

Hence $1+\sqrt{-3} \notin (2)$ and we can conclude. □ ✓

(1)

ii)

t.s. $I^2 = 2I$.

proof Let $a \in I^2$. then is a a a finite sum of elements of the form

$$(2a_i' + a_i''(1+\sqrt{-3}))^2 \quad a_i'', a_i' \in \mathbb{Z}.$$

|| NO! it's a finite sum of elements of the form $(2a' + (1+\sqrt{-3})a'')(2b' + (1+\sqrt{-3})b'')$ $a', a'', b', b'' \in \mathbb{Z}$

Thus we can ~~or~~ write ~~every sum~~ such elements as.

$$2(2a_i'^2 - a_i''^2) + 2(a_i' a_i''(1+\sqrt{-3})) \in 2I$$

it follows that $I^2 \subseteq 2I$

On the contraing it holds that

$2I \subseteq I \cdot I = I^2$ since $(2) \subseteq I$ ✓

(1)

So we can conclude. □